Helena McMonagle, Ph.D.

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PROFESSIONAL SUMMARY

Quantitative ecologist and marine biogeochemist with expertise in fish ecology, ecological modeling, biological oceanography, marine carbon dioxide removal, and U.S. federal ocean policy. Experienced in communicating complex, scientific concepts with a variety of audiences and stakeholders. Proficient in open and reproducible science practices in data analysis, statistical modeling, and presentation of results.

EDUCATION

University of Washington, Seattle, WA 2019 - 2025
PhD, Aquatic and Fishery Sciences; **GPA**: 3.96/4.00. Advisors: Dr. Tim Essington and Dr. Ray Hilborn. Teaching assistant for courses in Ecological Modeling and Marine Conservation and Fisheries Management. Relevant coursework: Advanced Ecological Modeling (including spatial population dynamics modeling and marine protected area simulations), Linear Regression Modeling for Ecological Data, Multivariate Statistics.

Wellesley College, Wellesley, MA 2012 - 2016
B.A. in Biological Sciences with Honors; **GPA**: 3.73/4.00. Relevant coursework: Marine Biology, Inorganic & Organic Chemistry, Limnology of Baikal, Climate Change & Society.

Sea Education Association, Woods Hole, MA Spring 2015
Marine Biodiversity and Conservation semester program. Relevant coursework: Biological Oceanography, Ocean Science, Public Policy. Co-led research on six-week research cruise.

WORK & SERVICE EXPERIENCE

Fisheries Biologist, *Ocean Associates Inc. (under contract to NOAA), Seattle, WA* 2026 – Present

- Led study evaluating the impact of potential future spatial closures for offshore wind energy infrastructure on fisheries stock assessment and associated uncertainty in reference points.

Postdoctoral Research Scholar, *Arizona State University (remote)* 2026 (federal grant terminated)

- Coordinated and co-wrote literature review of ocean iron fertilization, including monitoring, verification, reporting, durability, and environmental consequences and co-benefits.

Ocean Science and Policy Knauss Fellow*, National Science Foundation, Alexandria, VA* 2025 – 2026

- Reviewed research proposals, checked proposals for compliance, and summarized complex panel review information for scientists who submitted oceanography research proposals to the agency.
- Provided technical assistance to policy and communications teams on topics in ocean sciences.
- Co-led and moderated event about ocean science and impacts of climate change on ecosystems for over 300 attendees from non-profit organizations, Tribes, government agencies, and Congress.
- Drafted work plans and concise reports on ocean co-use, ocean observing for climate and weather forecasting, deep seabed mining, harmful algal blooms, coastal resilience, and other ocean topics.
- Authored news stories that spotlighted the impact of NSF-funded research for the public.
- Researched and prepared report for internal stakeholders about Congressional mandates and executive orders directing the agency's ocean science and policy directives.
- Presented at four public events for ~200 visitors on climate change mitigation, adaptation, and the ocean's role in the carbon cycle at the Smithsonian National Museum of Natural History.

NSF Graduate Research Fellow, *University of Washington,* Seattle, WA 2019 – 2024

- Led and managed oceanographic research in marine biogeochemistry and food web ecology.
- Reviewed literature and quantified the role of fishes and zooplankton in the biological carbon pump to inform carbon budgets, carbon sequestration estimates, and fisheries management.
- Co-taught course in Ecological Modeling, including population dynamics modeling using R.
- Reviewed the literature on deep seabed mining and deep-sea fishing, and authored paper for policy-makers on trade-offs between deep-sea ecosystem services and emerging ocean industries.
- Managed freely available, annotated, reproducible code for all projects on GitHub.
- Planned and led fish and zooplankton sampling and fish respiration rate experiments at sea.
- Coordinated annual outreach event, Aquatic Sciences Open House at University of Washington. Hosted ~500 attendees, including K-12 students and their parents, for interactive activities.
- Led workshop for 15-20 undergraduate and graduate students pursuing research grants.

Research Assistant, *Woods Hole Oceanographic Institution (WHOI),* Woods Hole, MA 2017 – 2019

- Analyzed fish growth rates using microscopy and image processing software and measured nutritional condition of Arctic cod using molecular biology techniques (Llopiz Lab).
- Mentored an undergraduate student in study on mesopelagic fish otoliths and published results.
- Collected fish and zooplankton at sea; conducted data analysis and visualization.
- Examined effects of toxicants on gene expression and epigenetics in fish and published results.
- Mentored three undergraduate students in molecular biology and research methods as a volunteer lab hand for the Marine Biodiversity and Conservation Course at Sea Education Association.

Teaching Assistant, *Marine Biological Laboratory*, Woods Hole, MA 2016 – 2017

- Co-taught biogeochemistry and microbial ecology for 20 undergraduate students.
- Led terrestrial and coastal marine field work and taught independent research skills.

PUBLICATIONS

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1. Sato, M., Erickson, Z.K., **McMonagle, H.**, and Llopiz, J.K. 2026. Mesoscale eddy structure drives depth-dependent aggregation of mesopelagic fishes. Limnology and Oceanography (L&O).
2. Sato, M., Erickson, Z.K., Drago, L., Maas, A.E., **McMonagle, H.**, Steinberg, D.K. Temporal dynamics of mesopelagic fishes within a mesoscale eddy: a Lagrangian perspective, L&O Letters.
3. **McMonagle, H.**, Llopiz, J.K., Maas, A.E., Steinberg, D.K., Govindarajan, A.F., Essington, T.E. (2024). Quantifying uncertainty in the contribution of mesopelagic fishes to the biological carbon pump in the Northeast Atlantic Ocean. ICES Journal of Marine Science 81 (10), 2037-2051.
4. Bucklin, A., Batta-Lona, P.G., Questel, J.M., Wojcicki, M., Wiebe, P.H., Llopiz, J.K., Glancy, S., Caiger, P.E., **McMonagle, H.**, Francolini, R., Govindarajan, A., Jech, J.M. and Thorrold, S.R. (2024). Metabarcoding and morphological analysis of diets of mesopelagic fishes in the NW Atlantic Slope Water. Frontiers in Marine Science 11, 1411996.
5. Martin, A.P., Dominguez, A.B., Baker, C.A., Baumas, C.M., Bisson, K.M., Cavan, E., Freilich, F., Galbraith, Galí, E.M., Henson, S., Kvale, K.F., Lemmen, C., Luo, J.Y., **McMonagle, H.,** de Melo Viríssimo, F. *et al*. (2024). When to add a new process to a model–and when not: a marine biogeochemical perspective. Ecological modeling 498, 110870.
6. **McMonagle, H.**, Llopiz, J.K., Hilborn, R., Essington, T.E. (2023). High uncertainty in fish bioenergetics impedes precision of fish-mediated carbon transport estimates into the ocean's twilight zone. Progress in Oceanography 217, 103078.
7. Bisson, K., **McMonagle, H.**, Iglesias, I., Halfter, S., Gallo, N. (2023). Five reasons to take the precautionary approach to deep-sea exploitation. Communications Earth & Environment 4 (1), 152.
8. Govindarajan, A.F., Llopiz, J.K., Caiger, P.E., Jech, J.M., Lavery, A.C., **McMonagle, H.**, Wiebe,

P.H. and Weifeng Z. (2023). Assessing mesopelagic fish diversity and diel vertical migration with environmental DNA. Frontiers in Marine Science 10, 1-13.

9. Quigley, L.A., Caiger, P.E., Govindarajan, A., **McMonagle, H.**, Jech, J.M., Lavery, A.C., Sosik, H.M., Llopiz, J.K. (2023). Otolith characterization and integrative species identification of adult mesopelagic fishes from the western North Atlantic Ocean. Frontiers in Marine Science 10, 1217779.
10. Aluru, N., Hallanger, I., Bjornsdatter, L., **McMonagle, H.**, Harju, M. (2021). Hepatic gene expression profiling in Atlantic Cod *(Gadus morhua)* liver after exposure to organophosphate flame retardants revealed altered cholesterol biosynthesis and lipid metabolism. Environmental Toxicology and Chemistry 40 (6), 1639-1648.
11. Koslow, J.A., **McMonagle, H.**, Watson, W. (2017). Influence of climate on the biodiversity and community structure of fishes in the southern California Current. Marine Ecology Progress Series 571, 193-206.

Manuscript in preparation and under review:

12. **McMonagle, H.,** Llopiz, J.K., Maas, A.E., Steinberg, D.K., Govindarajan, A.F., Essington, T.E. Refining predictions of respiration rates of mesopelagic fishes and zooplankton using respirometry and proxy measurements. In preparation.
13. Batta-Lona, P.G., You, V., Llopiz, J.K., **McMonagle, H**., Govindarajan, A., Lefebvre, L.S., Wiebe, P.H., Bucklin, A. DNA metabarcoding of mesopelagic fish diets in the NE Atlantic: insights and applications for deep-sea food webs and ecosystems. In review, ICES Journal of Marine Science.

GRANTS AND AWARDS

John A. Knauss Ocean Policy Fellowship (2025), NSF Graduate Research Fellowship (2021-2024), School of Aquatic and Fishery Sciences Graduate Student Fellowship (2019-2020), Ecological Society of America's Graduate Student Policy Award (2024), School of Aquatic and Fishery Sciences Symposium's Best Lightning Talk (2022) and People's Choice Award (2023).

INVITED TALKS

Panelist at Ocean Sciences Meeting, Glasgow, UK. 2026. Invited speaker on panel about science policy opportunities and other opportunities for scientists in government for early career scientists and students.

Panelist at Ecological Society of America Annual Meeting, Baltimore, MD. 2024. Invited speaker on panel about early career opportunities in science policy for undergraduate and graduate students.

Guest lecturer for Marine Conservation and Management, **University of Washington,** Seattle, WA. 2023. Lecture: "Mesopelagic fish and trade-offs in ecosystem services" for ~90 undergraduate students.

Guest lecturer in Biological Oceanography, Sea Education Association, Woods Hole, MA. 2021. Lecture: "Mesopelagic fish and trade-offs in ecosystem services" for ~20 undergraduate students.

Guest lecturer for Marine Ecology, **Stonehill College,** Easton, MA. 2018. Lecture: "A Glimpse into the Ocean's Twilight Zone". Presented interactive lecture on mesopelagic ecology for ~30 undergraduates.

SELECTED CONFERENCE PRESENTATIONS

Ocean Sciences Meeting, Glasgow, UK. 2026. "To what extent do respiration rate measurements constrain uncertainty in the contribution of fishes to the biological carbon pump?" Oral presentation.

Ecological Society of America, Baltimore, MD. 2024. "Quantifying uncertainty in the contribution of mesopelagic fishes to the biological carbon pump in the Northeast Atlantic Ocean." Oral presentation.

Ocean Twilight Zone Symposium, Woods Hole, MA. 2023. "Identifying and tackling uncertainty in fish-mediated carbon flux". Oral presentation and poster.

Effects of Climate Change on the World's Ocean, Bergen, Norway. 2024. "High uncertainty in fish-mediated carbon transport into the ocean's twilight zone." In-person, recorded presentation available at: https://www.youtube.com/watch?v=ODXyQZLSYyk&t=5712s.

SAFS Graduate Student Symposium, Seattle, WA. "Fish and the carbon cycle: do they matter?" Best Lightning Talk, 2022. "Using GitHub's Copilot to Write R Code Faster". People's Choice Award, 2023.

Ocean Sciences Meeting, New Orleans, LA. 2016. "Relationships between climate and biodiversity of fish assemblages in the southern California Current region". Oral presentation.

WORKSHOPS & MEETING LEADERSHIP

Mesopelagic Science to Action Workshop, Boston, MA. 2025. Invited participant in workshop to develop a roadmap for science to action pathways for management of the mesopelagic zone.

Frontiers in Ocean Sciences, **National Science Foundation (NSF)**, Alexandria, VA. 2025. Planned and moderated free, public event on research in climate science, ecosystem resilience, and natural hazards. Available at: https://www.nsf.gov/events/7th-annual-frontiers-ocean-sciences-symposium/2025-06-30

Bridging International Activity and Related Research Into the Twilight Zone (BIARRITZ), 2019, Southampton, UK. Participated in an international workshop advancing collaboration across major mesopelagic research programs and represented Woods Hole Oceanographic Institution.

ICES Annual Science Conference, 2021. Co-convened session on "Biomass, biodiversity, and ecosystem services in the mesopelagic zone" with Dr. Tom Langbehn and Dr. Peter Wiebe. Produced documents for continued collaboration based on individuals' research focus for use after the session.

SELECTED PUBLIC OUTREACH

Public dissertation defense, Seattle, WA. "Confronting uncertainty in the contribution of twilight zone fishes to the biological carbon pump." Available at: https://www.youtube.com/watch?v=gOk-dDIGUB4

University of Washington Scholars Studio, Seattle, WA. "What could deep-sea fishes (and their sinking poop) do for the ocean's ability to absorb our greenhouse gas emissions?" February 2022. Live-streamed oral presentation, available at: https://tinyurl.com/UWScholarsStudioMcMonagle

The Kenney Senior Living Community, Seattle, WA. "A Glimpse into the Twilight Zone". Spoke with about 50 community members about the ocean's twilight zone, ocean exploration, and the carbon cycle.

Falmouth High School, Falmouth, MA. "The Ocean Twilight Zone: Why Should We Care?" December, 2018. Interactive presentation at an optional assembly for high school students. Presentation published by Falmouth High School and available at: https://www.youtube.com/watch?v=N8QoDlCZpEI

SKILLS

Technical Skills: Highly proficient in ecological modeling, marine biogeochemistry, statistics, R, and MS Office. Introductory Python, MATLAB, and ArcGIS. Peer reviewer for four scientific journals.

Communication Skills: WHOI Media Training, NOAA Facilitation Basics Course, UW Science Communication Course, NSF Merit Review Basics Course on reviewing NSF research proposals. Proficient in Spanish. Excellent skills in oral and written communication and in building partnerships.

Field Experience: 2000+ hours of fieldwork in terrestrial, coastal, and pelagic ecosystems in Puerto Rico, Siberia, Cape Cod, Caribbean Sea, Bristol Bay, Slope Sea, and the Northeast Atlantic Ocean. Oceanographic data collection and sample processing (e.g., plankton nets, MOCNESS deployment, and CTD rosettes). Fish and zooplankton respirometry. Ecological time series collection. SCUBA Certified.